

What is the Effect of Synbiotic Therapy Using Lactobacillus Rhamnosus GG on Systemic Inflammatory Markers (IL-6, TNF- α) and Pulmonary Function (FEV1) in Patients with Stage 3 COPD?

Sydney Martinez, Keith Lee, Annapoorni Sanjeev, Adil Mukhi, and Jasmine Dong

STEMBuddies Canada

Zein and Monica

August 3rd, 2025

Word Count: 999 words

Abstract

Chronic Obstructive Pulmonary Disease (COPD) is a progressive lung condition marked by chronic inflammation and reduced airflow, currently ranking as the third leading cause of death in the United States. While conventional treatments emphasize medications and invasive procedures, emerging research highlights the gut-liver axis--a communication network between the intestinal microbiome and liver--as a critical factor in COPD progression. Disruption of this axis, often through gut barrier dysfunction and translocation of bacterial endotoxins like lipopolysaccharides (LPS), can trigger systemic inflammation and worsen pulmonary outcomes. This paper investigates the therapeutic potential of synbiotic intervention using *Lactobacillus rhamnosus* GG in Stage 3 COPD patients, with lung function measured via forced expiratory volume in one second (FEV1). Findings from existing literature suggest that *L. rhamnosus* GG may strengthen gut barrier integrity and modulate immune responses, leading to reduced inflammation and improved pulmonary health. Although individual variability in gut microbiota presents challenges, this approach offers a promising, non-invasive adjunct to standard COPD care. Further research is warranted to optimize dosage, assess long-term effects, and personalize treatment strategies. This study underscores the potential of targeting the gut-liver-lung axis as an innovative pathway for managing COPD.

Key words: Chronic Obstructive Pulmonary Disease, COPD, Gut microbiome, Gut-liver axis, *Lactobacillus rhamnosus* GG

Introduction

Chronic Obstructive Pulmonary Disease (COPD) is a progressive respiratory condition characterized by persistent inflammation and obstruction of airflow. It is the third leading cause of death in the US, affecting 13.5 million Americans (Sansom, 2014). While traditional treatment approaches focus on medication and invasive procedures, new evidence suggests a more systemic view, particularly involving the gut-liver axis. The gut-liver axis is the bidirectional communication between the intestinal microbiome and the liver, mediated by immune signaling and metabolic pathways. In COPD patients, an unhealthy gut and a weakened gut barrier (leaky gut), allow harmful bacterial substances like lipopolysaccharides (LPS) to pass into the liver. This activates the liver's immune system and leads to widespread inflammation, which can increase inflammation and pulmonary decline. To investigate this connection, this study will examine the effect of synbiotic therapy using *Lactobacillus rhamnosus* GG on lung function, specifically forced expiratory volume in one second (FEV1), in Stage 3 COPD patients. Studies have demonstrated the anti-inflammatory properties of *L. rhamnosus*, including its ability to reinforce gut barrier integrity and modulate cytokine signaling (Carvalho, et al, 2020). By focusing on the gut-liver-lung axis, this research explores a non-invasive intervention to target inflammation and improve pulmonary health. The objective of this paper is to evaluate the effect of *L. rhamnosus* GG on pulmonary function by modulating the gut-liver axis in Stage 3 COPD patients.

Hypothesis

If synbiotic therapy using *Lactobacillus rhamnosus* GG reduces inflammation through modulation of the gut-liver axis, then pulmonary function (measured by FEV1) in patients with stage 3 COPD will improve.

Rationale

Chronic Obstructive Pulmonary Disease (COPD) is a progressive respiratory condition that is characterized by long-term inflammation. Stage 3 COPD is marked by persistent airflow limitation and inflammation. This contributes to the disease's progression and reduced lung function, which is measured by FEV1. Studies have emphasized the gut-liver axis as a pathway linking intestinal imbalance in gut microbiota to increased inflammation due to an increase in gut permeability and the activation of the liver's immune system. With the intestinal barrier compromised, it allows bacterial byproducts such as lipopolysaccharides (LPS) to locate to the liver, activating an inflammatory response which can worsen lung function. *Lactobacillus rhamnosus* GG has been shown to restore gut barrier integrity, modulate immune system signaling, and reduce inflammatory markers such as TNF- α and IL-6, which are known to reduce lung function in COPD patients (Khailova, et al, 2014). Therefore, synbiotic therapy targeting the gut-liver axis can improve lung function. This paper evaluates how the presence of *Lactobacillus rhamnosus* GG influences inflammatory markers which can then influence

pulmonary function in patients with Stage 3 COPD. By placing focus on how the microbiome of the gut can affect pulmonary health, this paper aims to highlight a non-invasive treatment for patients with stage 3 COPD.

Experimental Design

Understanding how the effects of *Lactobacillus rhamnosus GG* influence COPD development is crucial. This study will use a randomized and placebo-controlled trial to investigate the effects of a synbiotic therapy on COPD patients. The independent variable is the administration of a daily synbiotic supplement containing *Lactobacillus rhamnosus GG* versus a placebo. The dependent variables are pulmonary function (FEV1) and serum inflammatory markers (IL-6 and TNF- α). Participants will be randomized into two groups, with the placebo group serving as a negative control to isolate the effect of the probiotic. This design allows for a clear comparison of outcomes between the treatment and control groups, which could help clarify the relationship between gut microbiome modulation and improvements in lung function.

Methodology

This study will employ a 12-week randomized and placebo controlled trial to investigate the synbiotic therapy on Stage 3 Chronic Obstructive Pulmonary Disease (COPD) patients. A total of 120 participants aged 50-75 and meeting GOLD guidelines for Stage 3 COPD, will be recruited and randomly assigned to either an experimental group receiving daily synbiotic supplement containing *Lactobacillus rhamnosus GG* and a prebiotic fiber, or a placebo group

receiving only the prebiotic fiber. The primary outcome measure will be Forced Expiratory Volume in 1 second (FEV1), assessed by spirometry, while secondary outcomes will measure serum levels of Interleukin-6 (IL-6) and Tumor Necrosis Factor-alpha (TNF- α) using ELISA. Statistical analysis will utilize a two sample t-test to compare the mean changes from baseline between two groups, with a p-value of < 0.05 considered significant.

Conclusion

The proposed study aims to investigate the effects of synbiotic therapy on COPD, and its design is strongly supported by a growing body of research. Pre-existing clinical and animal studies demonstrate that probiotics, particularly *Lactobacillus rhamnosus* GG, can delay acute exacerbations and reduce inflammatory responses in respiratory conditions. A pilot study conducted by Shanghai medical college in 2024 showed that probiotics significantly delayed the onset of escalations in patients with moderate-to-severe COPD (Hua et al., 2024). Furthermore, a study conducted in the University of Sao Paulo done with murine models demonstrated that pre-treatment with *Lactobacillus rhamnosus* had a protective effect against lung inflammation. The study showed a reduction in pro-inflammatory cells and cytokines, suggesting a mechanistic link between the probiotic and the mitigation of lung damage (Olimpio et al., 2021, Zuhair et al., 2025). Likewise, a systematic review conducted by the University of Sao Paulo concluded that probiotics can modulate the inflammatory process. The review specifically noted that some studies reported significant reductions in inflammatory markers such as IL-6 and TNF- α (Maia et al., 2019). This body of evidence is further supported by literature on the link between gut microbiome health and pulmonary function (Druszczyńska et al., 2024, Song et al., 2024). Therefore, the presented methodology is grounded in existing scientific evidence and seeks to

provide definitive clinical data on the efficacy of synbiotic therapy in managing Stage 3 COPD. Still, there are a few challenges, such as variation in gut bacteria among individuals. An additional factor to consider is that reaction to probiotics is not uniform and differs from individual to individual. Moving forward, more research is needed to test long-term effects, find the right dosages, and possibly tailor treatments based on individual gut profiles. In conclusion, this paper highlights the gut-liver axis as a viable solution for COPD, promising both as a supportive and potentially preventive strategy.

Works Cited

1. Carvalho, J. L., Miranda, M., Fialho, A. K., Castro-Faria-Neto, H., Anatriello, E., Keller, A. C., & Aimbire, F. (2020). Oral feeding with probiotic lactobacillus rhamnosus attenuates cigarette smoke-induced COPD in C57BL/6 mice: Relevance to inflammatory markers in human bronchial epithelial cells. Retrieved from <https://pmc.ncbi.nlm.nih.gov/articles/PMC7182195/>
2. Druszczyńska, M., Sadowska, B., Kulesza, J., Gąsienica-Gliwa, N., Kulesza, E., & Fol, M. (2024, November 15). The Intriguing Connection Between the Gut and Lung Microbiomes. *MDPI*, 13(11), 1005. <https://doi.org/10.3390/pathogens13111005>
3. Hua, J., Yang, Z., Cheng, Q.-J., Han, Y., Li, Z., Dai, R., He, B., Wu, Y., & Zhang, J. (2024, January 04). Prevention of exacerbation in patients with moderate-to-very severe COPD with the intent to modulate respiratory microbiome: a pilot prospective, multi-center, randomized controlled trial. *Frontiers*, 10. 10.3389
4. Khailova, L., Petrie, B., Baird, C. H., Dominguez Rieg, J. A., & Wischmeyer, P. E. (2014). Lactobacillus rhamnosus GG and Bifidobacterium Longum attenuate lung injury and inflammatory response in experimental sepsis. Retrieved from <https://pmc.ncbi.nlm.nih.gov/articles/PMC4022641/>
5. Maia, L. P., Yara Loyanne de Almeida Silva Levi, Y., Leal do Prado, R., & Santinoni, C. (2019, March). Effects of probiotic therapy on serum inflammatory markers: A systematic review and meta-analysis. *Journal of Functional Foods*, 54(9), 466-478. 10.1016/j.jff.2019.01.051
6. Olimpio, F., Mateus da Silva, J. R., Vieira, R. P., Oliveira, C. R., & Aimbire, F. (2021, November 05). Lactobacillus rhamnosus modulates the inflammatory response and the subsequent lung damage in a murine model of acute lung inflammation. *Elsevier*. 10.1016/j.clinsp.2022.100021
7. Sansom, W. (2014, November 13). *New COPD websites help patients, family members and health care providers learn best ways to treat chronic illness*. UT Health San Antonio.

<https://news.uthscsa.edu/new-copd-websites-help-patients-family-members-and-health-care-providers-learn-best-ways-to-treat-chronic-illness/>

8. Song, X., Chang, J., Dou, X., Zeng, X., Xu, Q., & Xu, C. (2024, October 24). The role and mechanism of gut-lung axis mediated bidirectional communication in the occurrence and development of chronic obstructive pulmonary disease. *PubMed*, 16(1).
10.1080/19490976.2024.2414805
9. Zuhair, M. N., Wiriansya, E. P., Masyhuri, I., Fakhri, M., Djaharuddin, I., & Bukhari, A. (2025, June 30). Role of *Lactobacillus rhamnosus* GG in the Management of Respiratory Diseases: A Systematic Review and Meta-Analysis. *Preventive Nutrition and Food Science*, 30(3), 222-229. <https://doi.org/10.3746/pnf.2025.30.3.222>

Appendix A: Definitions

COPD (Chronic Obstructive Pulmonary Disease): A chronic inflammatory lung disease that causes obstructed airflow from lungs.

Lactobacillus rhamnosus GG: A specific strain of probiotic bacteria that is commonly found in the gut.

Lipopolysaccharides (LPS): Large molecules found in the outer membrane of gram-negative bacteria. When these bacteria die, LPS can be released, triggering a strong inflammatory response in the host.

Synbiotic Therapy: A dietary supplement that contains both probiotics (beneficial bacteria) and prebiotics (a type of fiber that serves as food for probiotics).

Pulmonary Function: The measure of how well the lungs are working.

Cytokine: A small protein that is crucial in controlling the growth and activity of other immune system cells and blood cells. Cytokines are released by cells and affect the behavior of other cells.

FEV1 (Forced Expiratory Volume in 1 second): The maximum volume of air a person can exhale in the first second of a forced breath. It is a common measure of lung function.

GOLD Guidelines: A set of internationally recognized recommendations for the diagnosis, management, and prevention of Chronic Obstructive Pulmonary Disease (COPD).

Gut-Liver Axis: The bidirectional communication pathway between the gut microbiome and the liver, which influences metabolic function, immune responses, and inflammatory processes throughout the body.

Gut Microbiome: The community of microorganisms (bacteria, fungi and viruses alike) that live in the digestive tract.

Gut Microbiome Modulation: The process by which the composition or function of the microbiome is intentionally altered.

IL-6: A pro-inflammatory cytokine that plays a key role in the body's immune response and is often elevated in chronic inflammatory conditions like COPD.

TNF- α : A potent pro-inflammatory cytokine produced by the immune system, involved in systemic inflammation and cell death. High levels of TNF- α are associated with increased inflammation in COPD patients.